FlyByWire feature user manuals

One manual per `core/*` branch of the fork (36 branches) and one per fork-only `program/*` branch (2): what each feature does, how to use it, how to test it in the sim, what to expect, and its limitations with the reason

Aircraft	A32NX and A380X
Where to find it	This folder: each manual as HTML (with its `img/NN/` pictures) and as PDF; `FBW-user-manuals.pdf` holds them all, `FBW-user-manuals.docx` is the editable Word version
Branch	The `core/*` branches of the fork `TCdr/aircraft`: what to open the upstream pull requests from
Status	Restructured from the 31 per-PR manuals (their content is merged into the core manuals); each feature carries its own test status.
Pictures	Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams. Features added after the per-PR manuals are described in text.

1. The manuals
2. Program branches (fork-only, not upstream PRs)
3. Stacks: what must be merged first
4. Companion changes in aircraft-large-files
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1. The manuals

#	Manual	Aircraft	Branch	Based on	Features
1	Weather radar and terrain on the ND and VD (PDF)	A32NX and A380X	`core/nd-weather-terrain`	`master`	8
2	Surveillance: SURV controls, ECAM SURV lines and TCAS (PDF)	A380X (SURV panel, VD messages, ECAM lines), A32NX + A380X (TCAS)	`core/surveillance`	`master`	8
3	Independent CPT and F/O MCDUs (PDF)	A32NX	`core/a32nx-mcdu`	`master`	3
4	ECAM procedures, EMER CANC and FIRE TEST (PDF)	A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option)	`core/ecam-fws`	`master`	5
5	Runway awareness: RAAS and OANS (PDF)	A32NX (RAAS) and A380X (OANS)	`core/oans-btv-raas`	`master`	5
6	A380X standby instruments (ISIS) (PDF)	A380X	`core/isis`	`master`	4
7	Lights: strobe and beacon timing, interior lighting presets (PDF)	A32NX and A380X	`core/lights`	`master`	3

8	Fire warnings and the ENG FIRE pushbutton (PDF)	A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve)	`core/fire`	`master`	3
9	Electro-hydrostatic actuator accumulator start pressure (PDF)	A32NX and A380X	`core/eha-accumulator-start-pressure`	`master`	1
10	A380X MFD FMS and SURV pages (PDF)	A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit, the received company winds and the CHECK ALTN WIND check is also in the A32NX FMS; the A32NX keeps its own alternate cruise level and history wind rules; changes the A32NX MCDU too)	`core/a380x-mfd-fms`	`core/surveillance`	18
11	A380X ATC COM and datalink (PDF)	A380X (the ACARS provider connection is shared with the A32NX)	`core/a380x-atccom-datalink`	`core/a380x-mfd-fms`	7
12	A380X fuel jettison and FINAL fuel (PDF)	A380X	`core/fuel`	`core/a380x-mfd-fms`	7
13	GPS receivers, GPS deselection and ECAM STATUS (PDF)	A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only	`core/gps`	`core/a380x-mfd-fms`	8
14	A380X interactive ND (PDF)	A380X	`core/a380x-interactive-nd`	`core/a380x-mfd-fms`	2
15	flyPad performance calculators (PDF)	A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only)	`core/efb-performance`	`core/a380x-mfd-fms`	12
16	Cockpit printers (PDF)	A380X (printing, buttons) and A32NX (paper door, no torn paper edge)	`core/printer`	`core/a380x-atccom-datalink`	5
17	flyPad ground: taxi route, ATC frequencies and GSX (PDF)	A32NX and A380X (the OANS flags are A380X only)	`core/efb-ground`	`core/efb-performance`	8
18	flyPad Material 3 redesign and independent flyPads (PDF)	A32NX and A380X (the Buffet tab: A32NX only)	`core/efb-redesign`	`core/efb-ground`	27
19	A32NX air conditioning per the FCOM (PDF)	A32NX	`core/a32nx-air-conditioning`	`core/gps`	4

20	A380X cockpit buttons, lights and probes (PDF)	A380X	`core/a380x-cockpit`	`master`	6
21	A32NX exterior: spoilers by panel group and visible RAT (PDF)	A32NX	`core/a32nx-exterior`	`master`	2
22	A380X electrical alerts and AC ESS loads (PDF)	A380X	`core/a380x-elec`	`core/gps`	6
23	A380X flyPad failures (PDF)	A380X	`core/failures-a380x`	`master` + `core/fire` + `core/ecam-fws`	24
24	A32NX flyPad failures (PDF)	A32NX (the fire visual effects, the smoke detection function and the anti-ice systems part also A380X)	`core/failures-a32nx`	`core/gps` + `core/fire` + `core/fcom-audit-fixes` + `core/a32nx-air-conditioning`	50
25	Display unit failures and reconfiguration (PDF)	A32NX and A380X	`core/display-unit-failures`	`core/nd-weather-terrain`	6
26	A380X weather radar GAIN, ELEVN and TILT (PDF)	A380X	`core/a380x-wxr-controls`	`core/nd-weather-terrain` + `core/a380x-mfd-fms`	5
27	A380X VD airports as flat ground (PDF)	A380X (the map view fix: A32NX and A380X)	`core/a380x-vd-runways`	`core/nd-weather-terrain` + `core/oans-btv-raas`	3
28	FCOM audit fixes (PDF)	A32NX and A380X	`core/fcom-audit-fixes`	`master`	16
29	RAT ground stow and A32NX RAT performance (PDF)	A32NX and A380X	`core/rat-ground-stow`	`master`	2
30	A380X cargo doors follow the hydraulics (PDF)	A380X (the full-open fix: A380X and A32NX)	`core/a380x-cargo-doors`	`master`	2
31	A380X ATC mailbox with the SD on an ND display unit (PDF)	A380X	`core/a380x-mailbox-du-reconf`	`core/display-unit-failures` + `core/a380x-atccom-datalink`	2
32	A380X ELEC, fuel and FCOM gap alerts (PDF)	A380X (A32NX: ADIRU power supplies, engine anti-ice valves and anti-ice key events, through shared code)	`core/a380x-elec-fuel-alerts`	`core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes`	31
33	A380X OIS and ATC datalink system model (PDF)	A380X	`core/a380x-ois-datalink`	`master` + `core/printer` + `core/a380x-mailbox-du-reconf` + `core/a380x-elec`	8

34	A380X OIS: NSS AVNCS and FLT OPS applications (PDF)	A380X	`core/a380x-ois`	`master` + `core/a380x-ois-datalink` + `core/a380x-cockpit` + `core/efb-performance`	23
35	A380X COMPANY MSG memo and voice (PDF)	A380X	`core/a380x-ois-alerts`	`core/a380x-ois` + `core/a380x-elec-fuel-alerts`	1
36	A380X staggered cockpit power-up (PDF)	A380X	`core/a380x-power-up`	`core/a380x-ois` + `core/isis`	2

2. Program branches (fork-only, not upstream PRs)

These two branches are cut from `develop` of the fork and are not meant to be opened as upstream pull requests: they hold the long programs of the fork (flyPad failure details, and the FCOM gap fixes with the predictive terrain and obstacle alerting). Their numbers continue the list above.

#	Manual	Aircraft	Branch	Based on	Features
37	flyPad failure details (PDF)	A32NX and A380X	`program/efb-failure-details`	`develop` `9b2228143`	6
38	FCOM gap fixes (program overview) (PDF)	A32NX and A380X	`program/fcom-gap-fixes`	`develop` `9ad68d612`	10

3. Stacks: what must be merged first

Most branches are based directly on `master` (FlyByWire master 66021c244 merged in; `core/a380x-cargo-doors` starts from it) and can be merged in any order. The others are stacked: a branch needs every branch under it merged first.

Stack	Merge order
A380X MFD	`core/surveillance` > `core/a380x-mfd-fms` > then any of `core/a380x-atccom-datalink`, `core/fuel`, `core/gps`, `core/a380x-interactive-nd`, `core/efb-performance`
Printer	… `core/a380x-atccom-datalink` > `core/printer`
flyPad	… `core/efb-performance` > `core/efb-ground` > `core/efb-redesign`
GPS	… `core/gps` > `core/a32nx-air-conditioning`, `core/a380x-elec`, `core/failures-a32nx` (which also merges `core/fire`, `core/fcom-audit-fixes` and `core/a32nx-air-conditioning`: its A32NX FWC alerts build on their code)
A380X failures	`core/fire` + `core/ecam-fws` > `core/failures-a380x` (merged in for the engine failures)
A380X ELEC, FUEL and FCOM gap alerts	`core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes` > `core/a380x-elec-fuel-alerts` (the alerts need the ELEC failures of one and the FWS engine and fuel code of the other; its FCOM alert additions also extend the A380X door, A-ICE and NAV alert files of core/fcom-audit-fixes)
ATC mailbox	`core/display-unit-failures` + `core/a380x-atccom-datalink` > `core/a380x-mailbox-du-reconf` (the mailbox follows the DU reconfiguration of one and the ATC datalink of the other)
ND	`core/nd-weather-terrain` > `core/display-unit-failures`; `core/nd-weather-terrain` + `core/a380x-mfd-fms` > `core/a380x-wxr-controls`; `core/nd-weather-terrain` + `core/oans-btv-raas` > `core/a380x-vd-runways`

Power-up stagger	`core/a380x-ois` + `core/isis` > `core/a380x-power-up` (its OIT and SND changes build on the OIT of one and the SND of the other)
Program branches (on develop)	`program/efb-failure-details` (cut from develop `9b2228143`) and `program/fcom-gap-fixes` (cut from develop `9ad68d612`) need `develop`; they are not upstream pull request branches
Independent (on master)	`core/a32nx-mcdu`, `core/ecam-fws`, `core/oans-btv-raas`, `core/isis`, `core/lights`, `core/eha-accumulator-start-pressure`, `core/fire`, `core/a380x-cockpit`, `core/a32nx-exterior`, `core/fcom-audit-fixes`, `core/rat-ground-stow`, `core/a380x-cargo-doors`
Feature split over two branches	The RAT stow row of `core/efb-redesign` needs `core/rat-ground-stow` (the systems side): merge both for the feature to work; the visible RAT of `core/a32nx-exterior` goes back in only after that ground stow
Feature split over two branches (cargo doors)	The cargo door row of `core/efb-redesign` (Ground > Services) reads the inhibition that only `core/a380x-cargo-doors` writes: merge both for the row to show the new state
Same panel.cfg section	`core/display-unit-failures` (A32NX `DuMessage` gauge) and `core/efb-redesign` (A32NX F/O flyPad) both add `[VCockpit21]`: whichever is merged second renumbers its section to `[VCockpit22]`
Features split over two branches (FCOM alert additions)	A380X smoke detection: the SDF Rust and cargo smoke panel lights are on `core/failures-a32nx`, the SMOKE alerts on `core/a380x-elec-fuel-alerts`. A380X NSS smoke: the NSS zone and the SMOKE light of the NSS MASTER sw on `core/failures-a32nx`, the flyPad failures on `core/a380x-elec-fuel-alerts` (both carry the same failure mapping). Smart FIVE HUNDRED: the flyPad pin option on `core/efb-redesign`, the callout on `core/a380x-elec-fuel-alerts`. A32NX CHECK APPR SELECTION: the MCDU message on `core/gps`, the PFD FMA line on `core/failures-a32nx`. Merge both in each case. When `core/failures-a32nx` and `core/a380x-elec-fuel-alerts` meet: adirs.rs differs only by the IR 10 min alignment (keep it), keep the a320 navigation.rs of core/failures-a32nx; the anti-ice Rust is identical
Shared engine failure module	`core/failures-a32nx` and `core/failures-a380x` both carry the same fbw-common engine failure files (engine_failure.rs, ata.ts): whichever is merged second keeps one copy; in `failures/mod.rs` and the A380X `a380_systems_wasm` failure list keep the GPS entries of `core/gps` followed by one ATA 72 block. Both also carry the relight EGT watch of engine_start.rs (the A380X schedule sets 865 °C, the A32NX none)

4. Companion changes in aircraft-large-files

Some features change the 3D cockpit models, which live in the separate FlyByWire repository `aircraft-large-files`. Their companion commits are on the fork `TCdr/aircraft-large-files` and must be merged with the core branch.

Companion commit	What it adds	Core branch
`5563e31`	A32NX MCDU screens: both MCDUs on one texture (UV halves)	`core/a32nx-mcdu`
`36b4bd9`	A380X pedestal printer paper	`core/printer`
`27d6123`	A380X printer door, roller unit and paper slit	`core/printer`
`99237fb`	A32NX printer door	`core/printer`
`24f7b02`	A380X F/O flyPad screen (independent flyPads)	`core/efb-redesign`
`d3060ea`	A32NX F/O flyPad screen (independent flyPads)	`core/efb-redesign`
`8c14d21`	A380X printer ON light (green window)	`core/printer`
`fa6c5d1`	A380X fuel jettison plume at the outlets (stacked on 8c14d21)	`core/fuel`

`b8fb739`	A380X MFP (multi-function probe) vane animations	`core/a380x-cockpit`
`b790e8d`	A32NX DU reconfiguration: 17 overlay glasses on the EIS screens, DU_MESSAGE and DU_BLANK materials (stacked on 99237fb and d3060ea)	`core/display-unit-failures`

5. How to read them

- Each manual has one numbered section per **feature** of its branch. The grey line under the feature title gives the aircraft, where to find it and its **status** (sim-tested, or not, with the date).
- **What it is for**: the problem the change solves, in a few sentences.
- **How to use**: numbered steps for normal use.
- **How to test**: a table of steps a tester follows in the sim, each with the expected result. If the sim does something else, it is a bug worth reporting with the branch name.
- **What you should see** tables: the expected behaviour in each situation; **Limitations and why**: what does not work yet, the reason and what to do about it.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.